Matthew L. McCoy

Additive Manufacturing Technical Lead | Machine Architecture & Process Development | Product Launch & Scale
NSF Graduate Research Fellow | Atlanta, GA

630-740-9348 mail@matthewmccoy.info www.matthewmccoy.info linkedin.com/in/matthewlmccoy

I build and architect industrial powder-bed additive manufacturing systems from first principles and lead multidisciplinary engineering teams that turn them into reliable, scalable products. At Formlabs, the world's largest supplier of professional SLA and SLS systems, I became a printer-wide technical authority in SLS R&D across materials, process physics, optics, thermals, inert-gas handling, hardware, controls, calibration, and software. I held engineering sign-off authority for the 30+ engineer Fuse 1+ 30W launch, owning process architecture, requirements, verification, factory software, and machine acceptance; the Fuse 1/1+ family now represents over 55% of SLS printers sold worldwide. I then returned to lead early Fuse X1 optical and thermal architecture. For my Ph.D., I built the first working platform for Radio Frequency Additive Manufacturing (RFAM), a volumetric polymer powder-bed process that sinters a whole part at once instead of rastering it, together with its multiphysics design software and fully matched RF heating network. **I bring the technical depth to make system-level decisions and the leadership to drive complex printer programs through production.**

SELECTED PROOF

- **Engineering authority:** led technical execution for a 30+ engineer program and 130+ ECOs; owned engineering sign-off for launch gates, factory acceptance, and shipment release on the fastest product launch in Formlabs history.
- **Printer-wide technical depth:** developed SLS materials, process physics, optics, thermal and inert-gas systems, hardware, controls, calibration, and factory and customer software.
- **First-principles range:** built a novel powder-bed process's machine, multiphysics design software, and RF power-delivery network from scratch, entering RF engineering with no prior domain background.

EXPERIENCE

Formlabs — May–Sep. 2019; May 2021 – Sep. 2023

Selective Laser Sintering Research & Development — *Somerville, MA*

Senior Optical and Thermal Systems Engineer (P3) — *Apr. 2023 – Sep. 2023*

- Owned early Fuse X1 optical architecture; reduced the laser, spot-size, hatch, material, smoke, and thermal design space to a few characterized options and presented the system recommendation to engineering leadership and the C-suite.
- Translated process physics into requirements, verification, and executive trade studies; contributed patent disclosures, mentored and hired engineers, and was later offered technical leadership of a five-engineer product-incubation team.

Print Process Engineer (Core R&D) & Technical Program Manager (P2) — *May 2021 – Apr. 2023*

- Engineering authority for the 30+ engineer Fuse 1+ 30W redesign and 130+ ECOs; approved EVT/DVT/PVT gates and production release in nine months. Fuse 1/1+ now represents over 55% of SLS printers sold worldwide.
- Owned process architecture, requirements, verification, factory-test software, and acceptance of contract-manufactured machines in China; trained factory teams, debugged launch issues across time zones, and held shipment-release authority.
- Developed SLS materials, process parameters and controls, upgraded thermal architecture, and a recirculating inert-gas system; created calibration tools that cut dimensional error 3× (±300 to ±100 μm) and enabled LEGO's global Fuse rollout.

Mechanical Engineering & Print Process Intern, Fuse SLS — *May 2019 – Sep. 2019*

- Resolved a laser-coordinate failure that had become a critical beta-testing bottleneck for part fitment; improved pre-launch Fuse geometry accuracy 25% through laser and calibration studies.
- Designed core Fuse Sift powder-processing functions and built the first working Fuse Blast automated media-blasting prototype, selected for later commercialization.

Georgia Institute of Technology – RFAM Platform Architect & Lead Developer — Sep. 2023 – Present

Ph.D. Candidate, Digital Design and Manufacturing Lab / NSF GRFP — *Atlanta, GA*

- Built the first working RFAM platform, a volumetric polymer powder-bed process that patterns conductive dopant and sinters the entire part at once with RF energy; secured ~$150K internal funding plus a ~$170K NSF Graduate Research Fellowship, and owned machine architecture, powder handling, high-viscosity jetting, controls, instrumentation, and safety systems.
- Designed and built the 13.56 MHz high-voltage RF power network from scratch: custom-wound ferrite step-up transformers, HT50 ceramic capacitor banks, impedance matching, and a Faraday enclosure. Reaching a stable match into a capacitive, lossy, continuously shifting load was non-trivial; converting that match into efficient energy transfer through the powder bed was harder still, and took 38 physical network builds guided by a fitted physics model that matched hardware within 2–3% and was released as open-source software.
- Melted and densified a graphite-doped nylon cube in under ten minutes, against the hours a conventional laser powder-bed fusion cycle requires; sustained 300 W at 0.17% reflected energy across a 138.5 kJ run while holding the match by hand as the load drifted through melt.
- Root-caused a common-mode RF leakage path corrupting in-situ thermography, cutting coax common-mode current 25 dB and restoring the camera to its RF-off noise floor at full power.
- Wrote and finite-difference-verified HEATR (High-frequency Electrothermal Additive Thermal Resolver), a custom solver coupling electrothermal, phase-change, and densification physics, with an adjoint pipeline that converts part geometry into filtered, 4-bit printable material maps; graded designs outperformed uniform material on every benchmark geometry and beat the best prior compensation method in 72% of cases.
- Co-authored peer-reviewed WAAM and hybrid WAAM/powder-blown DED research, including a 2025 *npj Advanced Manufacturing* paper; three journal manuscripts in preparation.

EDUCATION

Georgia Tech Ph.D. Mechanical Engineering | Minor: Industrial Design | NSF Graduate Research Fellow — **Sep. 2023 – Dec. 2026**

Dissertation: Advancing Volumetric Additive Manufacturing via Radio Frequency Energy Delivery: Process Development & Characterization

Northern Illinois University B.S. Mechanical Engineering | Summa Cum Laude — **Aug. 2017 – May 2021**

Minor: Music | GPA: 3.95/4.00 | DeKalb, IL